

DIYA =>

Good morning everyone, Today me and my friends are going to give a presentation on RAG, *click*

RAG also Retrieval augmented generation

click

What we will discuss today:

1st we will give an introduction to rag

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2nd we will talk about the RAG pipeline which means how rag works

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3rd we will talk about the steps to implement rag.

Click

4th we will give you a live demo

Click

Lets dive into the theory section

RAG is an AI framework that connects LLM(like chatgpt) to external, private knowledge. Like our notes and companies internal data.

So instead of just relying on what the ai learned during its initial training, RAG lets the model to look up real-time information to provide an accurate context aware answer.

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Why do we even need RAG? Lets discuss the problems, 1st despite how the LLM have improved they don't have any knowledge on our personal data, so they just generate generic data or it will just hallucinate.

And then their comes the idea of Zero-Shot learning, even if we upload the data the model don't exactly have the live updates on the data.

Now my friend will tell how RAG helps here.

(New person comes)

Click

Thank you diya,

Now I will tell you how rag helps

1ST by fixing the company data gap, RAG searches your private database for the relevant document snippets and attaches them to the prompt, which forces the llm to answer using the verified facts of our data instead of generic guess.\

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2nd by enabling Zero-shot live updates, this is done by separating ai's reasoning from storage, you update the connected database rather than retraining the entire model, Then LLM reads the fresh information. This also helps in saving the Training costs.

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Now lets discuss the RAG pipeline.

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This starts with document injection, here private documents are broken into text chunks and converted to numerical vector embeddings and then stored in the database.

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Then comes information retrieval, here when user asks the question, the system searches the database to find and pull the text chunks, that has the relevant facts.

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Then comes response generation, the system pins the retrieved facts directly to the user's prompt and hands it to the llm, where the llm generate the response.

(New person comes)

Thank you Deepika,

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Now lets start with Information extraction process

This starts with Loading the data we want to apply RAG.

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Then we will split the loaded data into chunks because of token limits and improving the search speed of the retriever, and give proper context.

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Then we will embed the data, giving it numerical representation, because computer cant understand the text directly.

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Then we will store the data into the vector database(vector database stores the data in vector(mathematical) form)

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Now lets talk about the information retrieval process

This starts with the user asking the question, defining the exact info that needs to be found.

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Then the question is also embedded and then compared to the document inside the database.

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Then comes the retrieval, the system searches the database for the exact text chunks that match the user's question, and then it appends them to the prompt with the exact context.

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Now lets finally get into the answer generation process, so we have prepared a simple chart for it.

This starts with the user

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User's question is then sent to vectorDB and prompt

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Then retriever takes the proper context from the DB and then gives those chunks to the prompt, where inside the prompt the user's query is added with the context.

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Then the generated prompt is sent to the LLM and it gives response to the user.